Copyright Protection with EXIF Metadata and Error Control Codes

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Abstract

In this paper, we propose a practical scheme for copyright protection of images with the aid of EXIF metadata and error control codes. Application for robust watermarking is one of the major branches in digital rights management (DRM) systems and digital forensics. By use of robust watermarking, it generally alters selected coefficients in the transform domain to accomplish the embedding process. We consider not only the image contents itself, but we also employ the EXIF metadata, which serves as the role of watermark, to be integrated into our scheme. Moreover, for enhancing the copyright protection capability, channel coding is employed. Taking the manufacturer, camera model, date and time stamp, and some other information in the EXIF metadata into account, conventional watermarking techniques can be applied to ordinary pictures taken by ourselves, and watermarked images with good quality can be produced. Even when the marked image has been intentionally modified, the original EXIF with selected information can mostly be recovered from the channel decoding process. Simulation results present the effectiveness of such an implementation.

1. Introduction

With the fact of the proliferation of consumer electronics devices, most people can easily produce his or her own digital pictures by using digital cameras at any time. As a result, digital images are being accumulated rapidly, and automated tools for organizing these pictures have become a necessity for users. With the large amount of pictures captured, in order to ease the task for organizing and classifying the vast amount of digital pictures, the EXchangeable Image File format (EXIF) is used to help store some information about the digital camera, including the date and time information, the settings of the camera, and the copyright information. In addition to the main purpose of EXIF for helping users to organize and classify the pictures taken, we can also apply the EXIF to the scope of digital forensics area to conquer the tampering problem that are frequently encountered due to the ease of editing the digital files. Therefore, we focus on watermarking with EXIF and BCH codes in this paper.

Here we use the digital images, taken by different cameras, to perform the applicability of watermarking with the aid of EXIF. It is generally agreed that for watermarking algorithms, the watermarked image quality (or *imperceptibility*), the survivability after intentional attacks (or *robustness*), and the number of bits embedded (or *capacity*) are the three most important factors to assess how good the algorithm is. Hence, we implement our algorithm with discrete cosine transform (DCT), choose the reasonable amount of information in EXIF metadata, and select the appropriate DCT coefficients for watermark embedding. To enhance the robustness of the extracted EXIF metadata, BCH codes are used, and better results can be expected. Experiments are conducted under the scenario that when the watermarked images are attacked by some means, the selected information in EXIF can be recovered back to some extent, and hence the copyrights of such images can be protected.

This paper is organized as follows. In Sec. 2 we describe EXIF and BCH codes for watermarking. In Sec. 3 we present the watermarking algorithm. Simulations are depicted in Sec. 4, and we conclude the paper in Sec. 5.

2. Watermarking with EXIF

2.1 Description of EXIF metadata

EXIF is a specification for the JPEG- and TIFF-image formats, mainly used by digital cameras [1][2]. It contains information about camera settings and shooting environment of the camera and the picture itself, including the date and time that the picture is taken, the camera manufacturer and camera model, the horizontal and vertical resolutions of the picture, shutter speed, ISO speed, white balance, flash utilization, etc. The copyright and geographic information can also be included [3][4]. EXIF metadata offer the utility for accurate searching, retrieval, and viewing purposes [5][6]. Unfortunately, they are

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usually not completely recorded due to different implementations and incomplete support of early digital cameras. Therefore, for using the metadata to serve as the watermark, the necessary portion should be carefully chosen. Figure 1(a) is an example for the picture taken by ourselves, and Figure 1(b) is the associated EXIF metadata extracted by our program. Taking EXIF utilities into account, we extend its scope to copyright protection and tamper resistance.

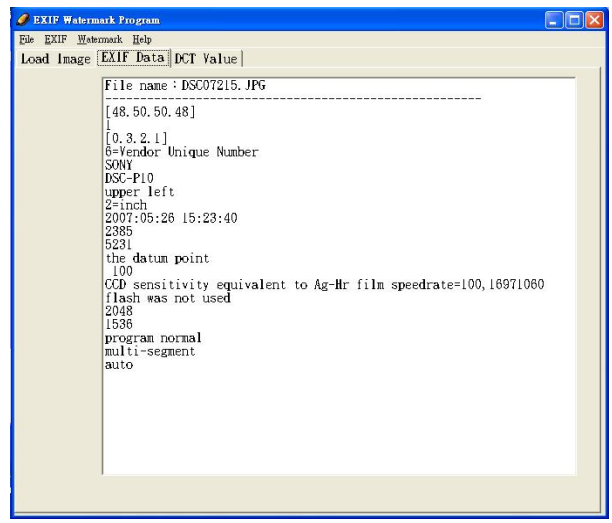
As we stated in Sec. 1, three requirements considered mostly are imperceptibility, robustness, and capacity. Taking practical implementations into account, we convert the EXIF metadata into bitstreams, and choose to embed 1 bit per 8×8 block in the middle frequency coefficients with different embedding strengths.

2.2 BCH-encoded watermark information

In this paper, we use the ordinary pictures taken by ourselves, illustrated in Figure 1(a), with the size of 2048×1536. Hence, with the setting that for embedding at most 1 bit per block, we are able to embed $\frac{2048}{8}\times\frac{1536}{8}=$ 49152 bits. We choose the information including camera model and vendor, time stamp, and other settings shown in Figure 1(b), which makes a total of 171 bytes. Since all the characters are represented by ASCII codes, we take the binary form of the information, 1368 bits in total, to serve as the watermark information. In order to enhance the robustness against attacks, unlike previous approaches to repeat the same watermark information for several times [8], channel coding can be employed for enhanced robustness [9]. We use BCH codes with different error correcting capabilities to encode the watermark information, and to generate the embedded bitstream.

We consider to use three BCH codes with different error correcting capabilities and redundancies, namely, (7, 4), (63, 45), and (511, 493) BCH codes. The watermark information is 1368 bits, and considering practical implementation, the numbers of bits for the watermark after BCH encoding are 2394, 1953, and 1533 bits, respectively, which are below the maximum number of allowable capacity. On the one hand, by use of BCH codes with more capability for error protection, more redundancy needs to be produced; hence more bits need to be embedded. On the other hand, fewer bits embedded means less alteration to the original image, and it leads to better watermarked image quality subjectively. We need to mention that we are going to demonstrate the applicability to use EXIF for watermarking. Since the numbers of bits for embedded bitstreams are far below the allowable capacity, more information can be also included whenever necessary.

(a)



(b)

Figure 1. (a) Original image in this paper. (b) Selected information of EXIF.

3. Proposed Algorithm

We follow conventional DCT-based watermarking techniques [10] with modifications for integrating with the EXIF metadata. Steps are depicted as follows.

3.1 Encoding EXIF with BCH codes

With previous experience, embedding of the repetition of watermarks is applicable. However, this method induced too many bits for embedding, and it caused the degradation of watermarked image. To alleviate this effect, we take error control codes into consideration, and we choose three BCH codes with different error correcting capabilities, namely, the (7, 4), (63, 45), and (511, 493) BCH codes. They lead to different redundancies, namely,

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75%, 40%, and 3.65%, respectively. Since the encoded bitstream must be below the maximum number of embedded bits, we can choose more data in EXIF and then encoded with fewer redundancy BCH code in error correction, and vice versa.

3.2 Information embedding with DCT

We modify some conventional scheme [10] for DCT-based watermarking to embed the binary watermark.

Step 1. *Performing the DCT of original image*: 8×8 DCT is performed on the entire 2048×1536 image. Since the original image is color image, we embed the watermark into the red plane. In one block, it leads to one DC coefficient, presented by position 0, and 63 AC coefficients, presented by positions 1 to 63.

Step 2. *Choosing the middle-frequency band:* It is suggested to embed into middle frequency coefficients [10] to meet the watermarking requirements. Therefore, we choose to embed BCH-encoded information into the AC11 coefficient of every block.

Step 3. *Obtaining the threshold for embedding:* The ratio between the average value of DC and AC11 coefficients among the $\frac{2048}{8} \times \frac{1536}{8} = 49152$ blocks is served as the threshold for watermark embedding.

Step 4. *Embedding the watermark:* By taking the threshold in Step 3, it meets either situation below.

- If bit 0 is embedded, and if the original AC11 is larger than the threshold, it is decreased to be smaller than threshold by a parameter δ, called embedding strength. If not, keep it unchanged.
- If bit 1 is embedded, the contrary to the previous situation is performed.

Step 5. *Performing the inverse DCT of modified coefficients:* Inverse DCT is calculated to obtain the watermarked image. We use the peak signal-to-noise ratio (PSNR) to evaluate the watermarked image quality and to determine whether the performance is acceptable or not.

3.3 Applying proper attacks and data extraction

For verifying our implementation, we choose JPEG compression, low-pass filtering (LPF), and median filtering (MF), to perform the attacks. Watermark bits can be extracted based on the threshold in Step 3 in Sec. 3.2. To assess the robustness, the bit-correct rate (BCR) before and after BCH decoding, and hence the decoded symbol-correct rate (SCR) values after experiencing these attacks are calculated. Here, the BCR's are the percentage of the number of correctly extracted bits, and the SCR is that of the correctly decoded ASCII symbol in EXIF metadata. The higher the two values, the better the outcomes.

4. Experimental Results

We choose the ordinary picture in Figure 1(a), the selected EXIF information in Figure 1(b), and perform watermark embedding with different embedding strengths into the AC11 coefficients in the DCT domain. The watermarked image is illustrated in Figure 2, with the PSNR of 47.07 dB.

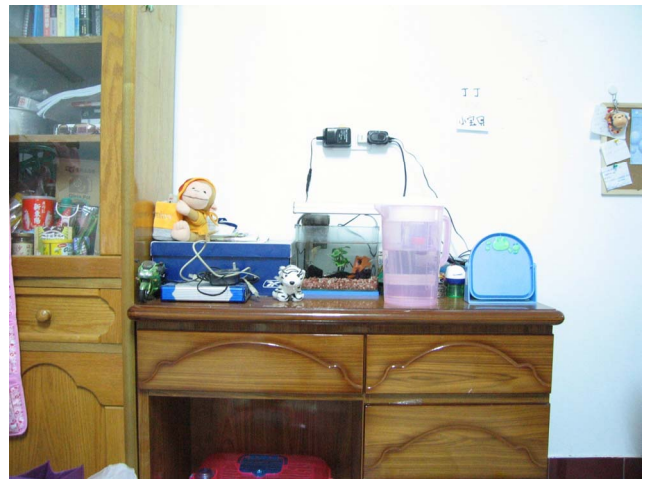

Figure 2. The watermarked image with PSNR = 47.07 dB.

We also apply JPEG, LPF, and MF attacks depicted in Sec. 3.3 for verifying the robustness in the EXIF metadata. By adjusting the watermarked image quality, we present the extracted BCR before and after BCH decoding in Figure 3(a) and (b), and SCR in Figure 3(c), respectively. We need to mention that when no attack is applied, all the bits can be extracted correctly, hence BCR = SCR = 100%. We only show the results under attacks.

Comparing Figure 3(a) and (b), we find that high BCR values can be obtained with somewhat smaller PSNR values, and deteriorated BCR values correspond to better PSNR values. Hence, there must be a trade-off for adjusting good BCR values with acceptable image quality. Also, BCR values in Figure 3(b) are higher than those in Figure 3(a), and this proves the error correcting capability with the use of error control codes.

Finally, in Figure 3(c), we observe that the SCR values are inferior to the corresponding BCR values. This is because we treat every 8 consecutive bits as a symbol, hence, if one bit is wrongly decoded, the symbol is consequently wrong. How to enhance the SCR values can be studied in the future.

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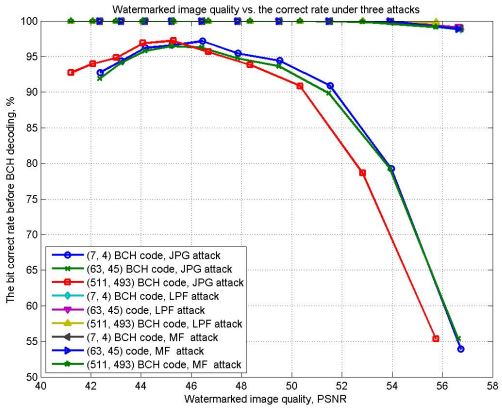
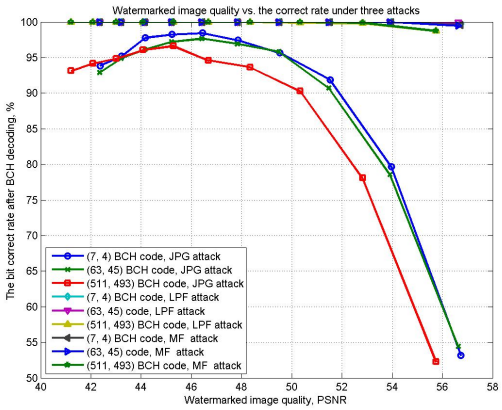
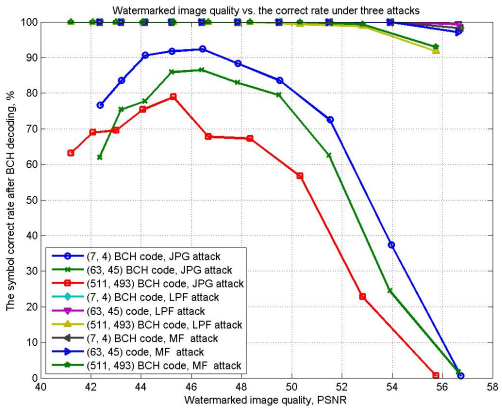


(a) Correct bit rates before BCH decoding



(b) Correct bit rates after BCH decoding



(c) Correct symbol rates after BCH decoding

Figure 3. Correct rates for extracted data under different image qualities. (a) BCR before BCH decoding. (b) BCR after BCH decoding. (c) The symbol correct rate.

Summing up, considering both the watermarked image quality and the robustness of the watermark, and comparing to the embedded capacity in literature, the number of embedded bits is reasonable. When we embed the EXIF metadata with reasonable capacity into the middle frequency coefficients in the DCT domain, acceptable image quality and robustness can be reached.

5. Conclusion

In this paper, we discussed about the practical implementation of robust watermarking with EXIF metadata and channel coding. By finding the necessary information in EXIF, it can be served as binary watermark, and further protected by BCH codes, for embedding. Simulation results depict that the metadata can be perfectly extracted with the sacrifice of watermarked image quality, and our algorithm is robust to common attacks. Hence, some trade-off needs to be searched for. The proposed scheme is suitable for JPEG-like image processing systems, and it has the potential for integrating the algorithm into consumer products.

6. References

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